Combining Machine-Learning and Dynamic Network Models to Improve Sepsis Prediction

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Supervisors: Prof. Dr. Tobias Knopp and M. Sc. Artyom Tsanda

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1. Sepsis
2. Dynamic Network Model
3. Latent Dynamics Model
4. Experimental Results
5. Conclusion

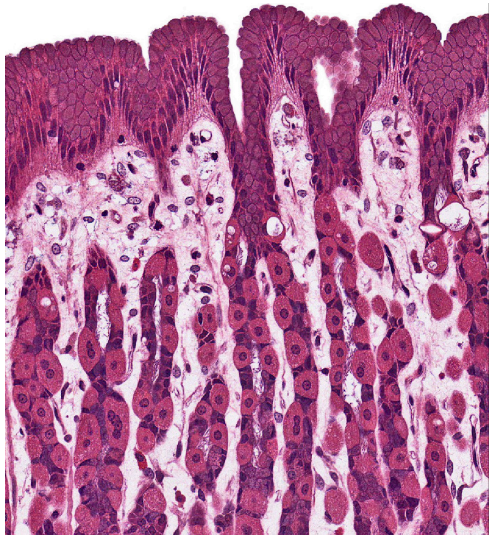
Sepsis

Sepsis

Multi-Scale

- Innate immune activation
- Cytokine-Storms
- Metabolic Failure
- Organ Dysfunction

Stomach tissue (400x magnification), parenchymal cells on top and stroma cells in light-pink. (taken from [1])



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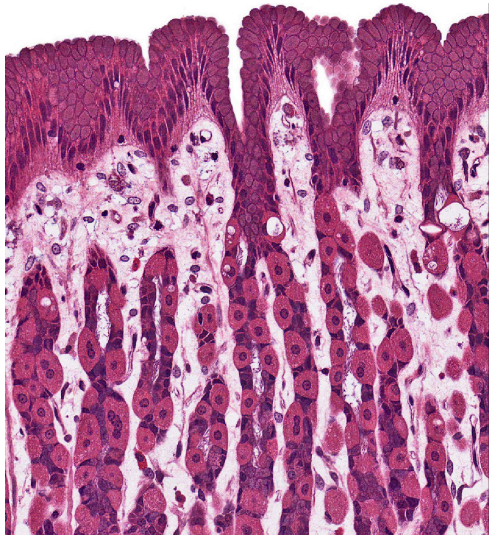

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High mortality and rapid progression

- ~50 million yearly patients
- ~11 million yearly deaths

System-wide damage

Long-term consequences

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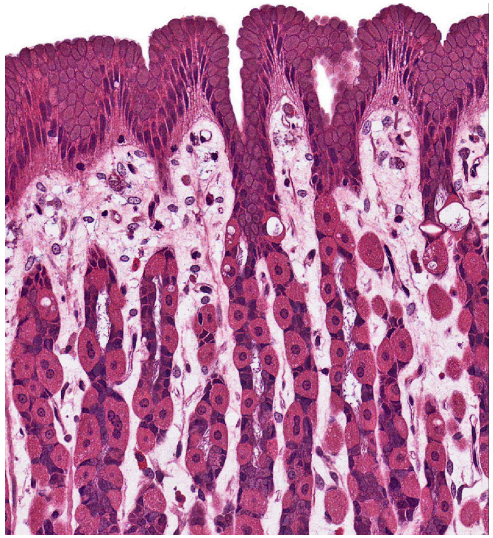

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Long-term consequences

Difficult to diagnose

Typical Deep Learning Problem

Dynamic Network Model

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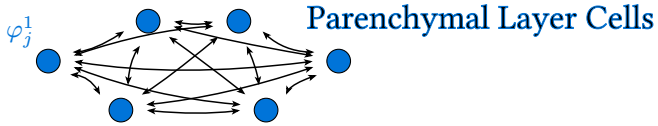
Dynamic Network Model (DNM)



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$$\dot{\kappa}_{ij}^1 = -0.03\left(\kappa_{ij}^1+\sin\left(\varphi_i^1-\varphi_j^1-\beta\right)\right)$$

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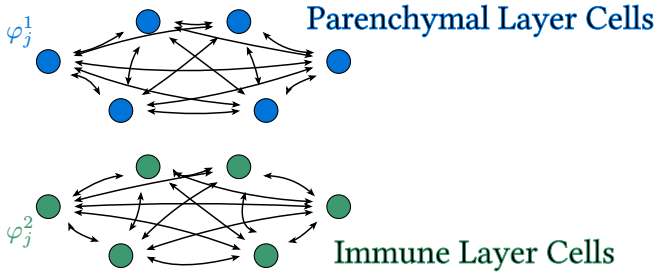

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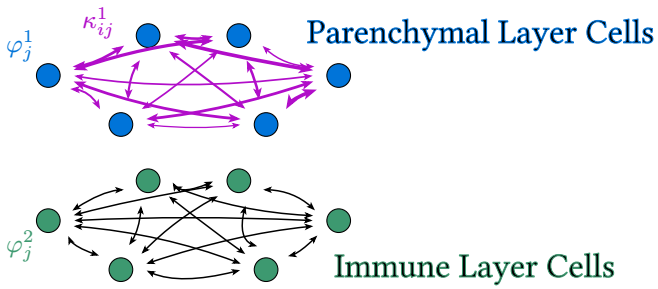

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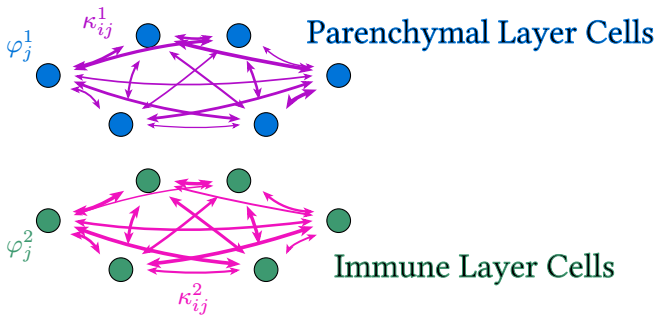

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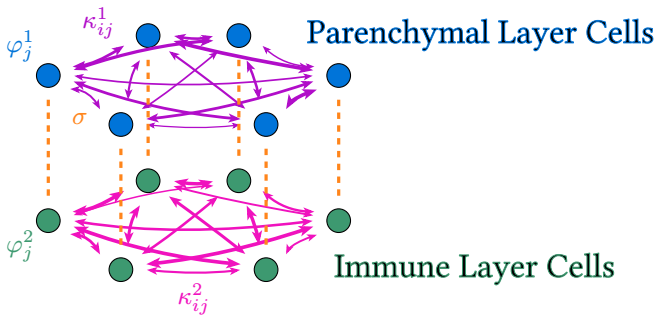

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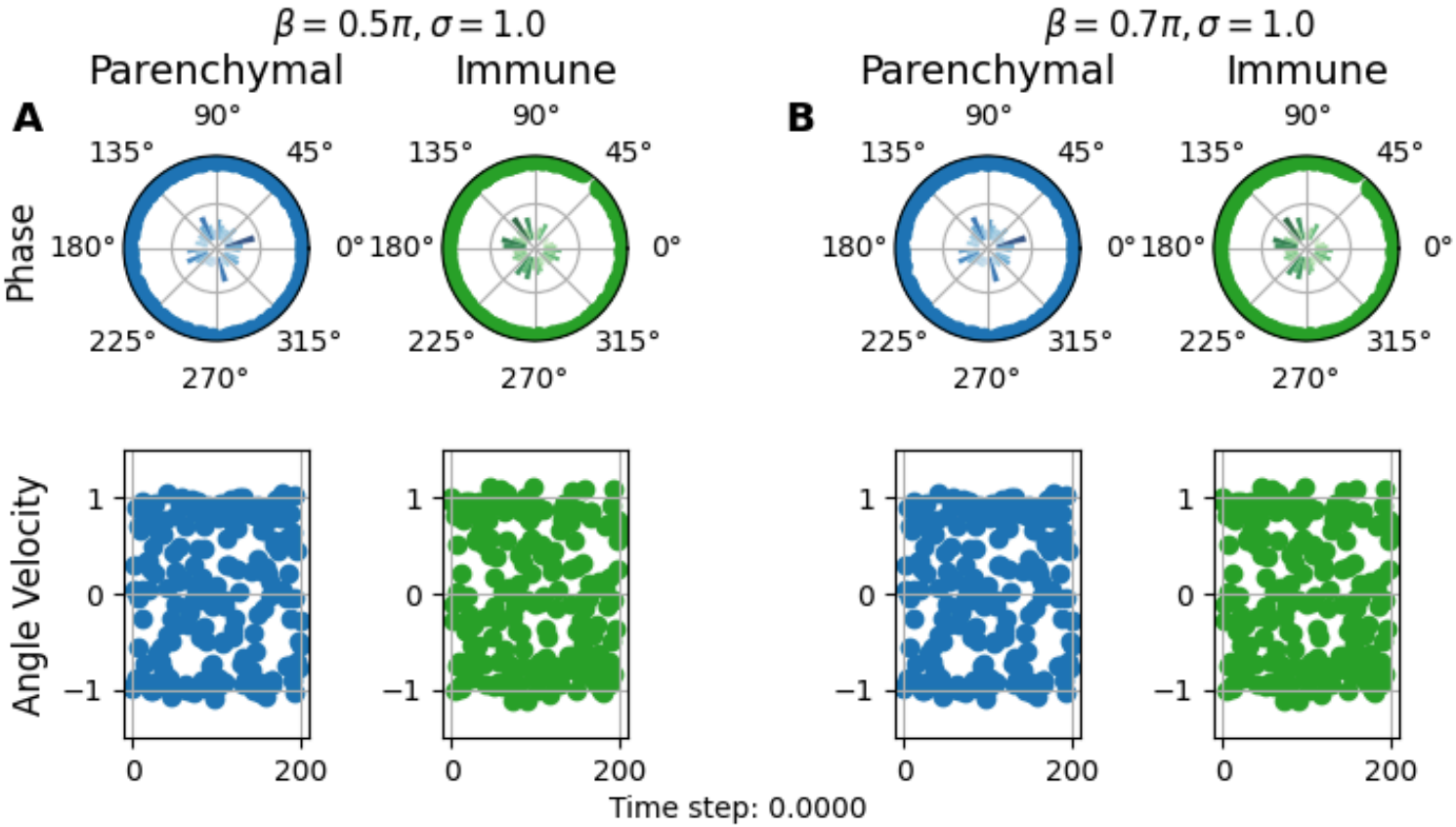

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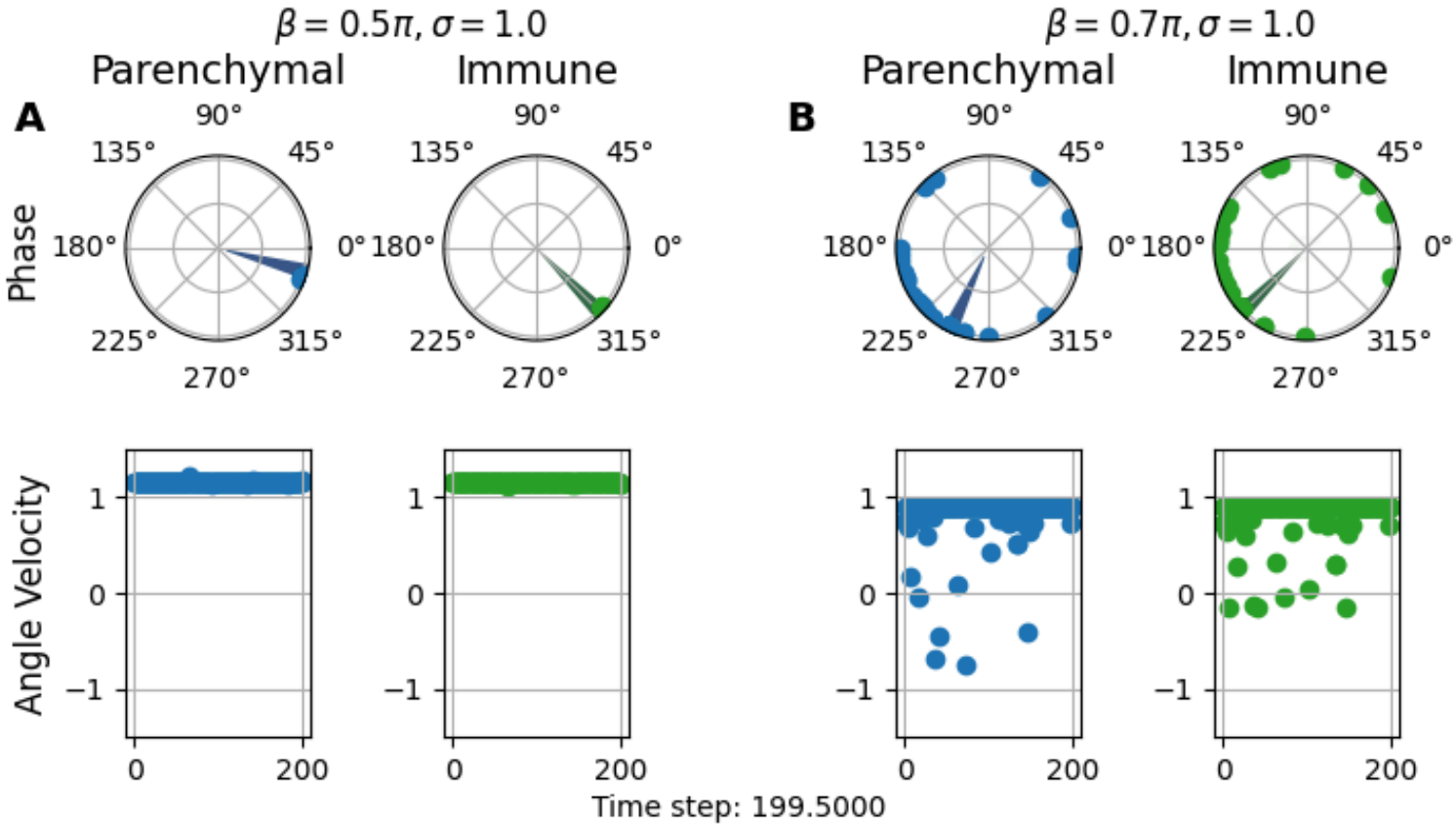

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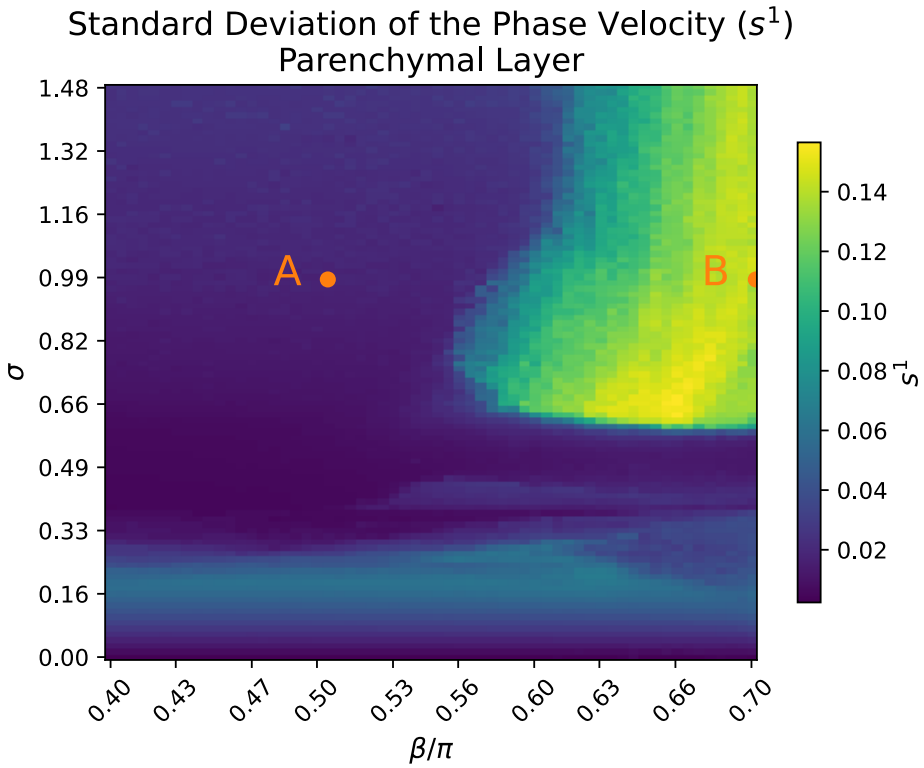

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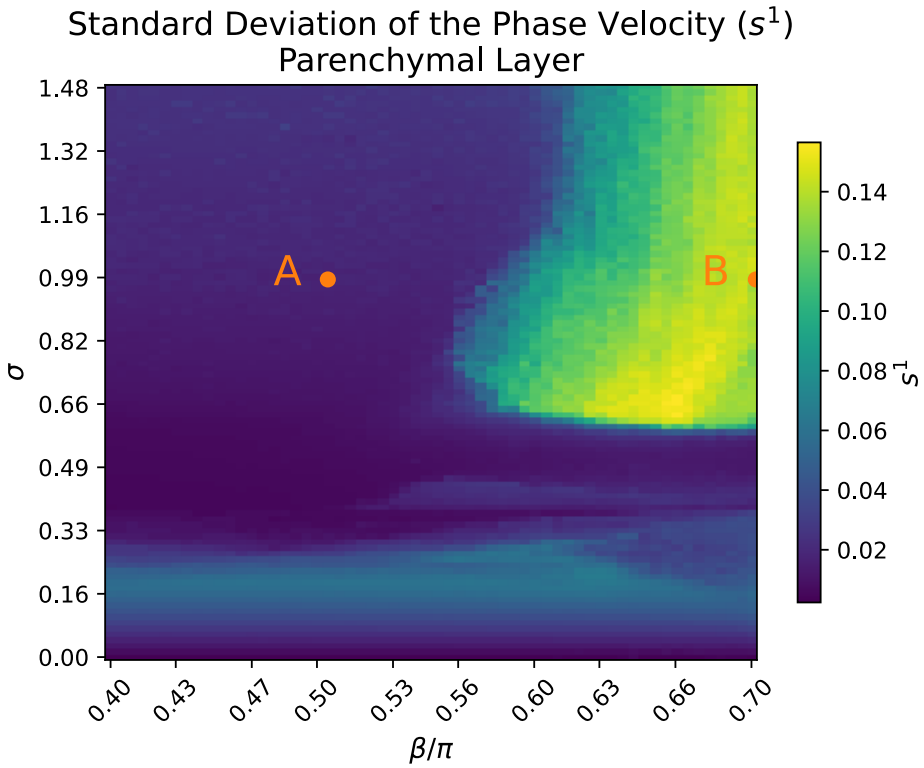


Summary

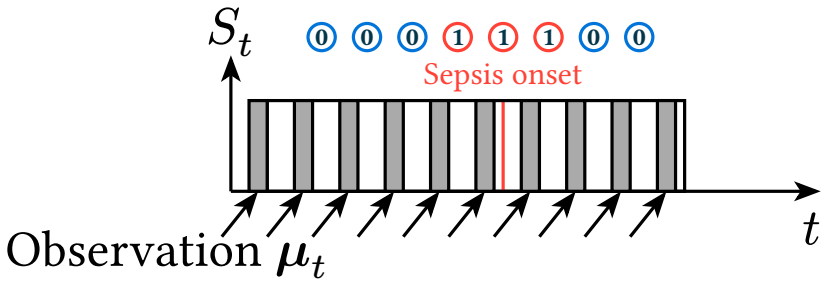
- Functional Model of critical transitions
- β and σ modulate synchronization
- Not yet validated on clinical data

Can we utilize it to overcome black-box predictions?

Latent Dynamics Model

Latent Dynamics Model

Online Prediction



Sepsis Definition $\rightarrow S_t$

1. Suspected Infection $\rightarrow I_t$
2. Dysregulated Host Response $\rightarrow A_t$
   - SOFA-score increase ≥ 2

Latent Dynamics Model

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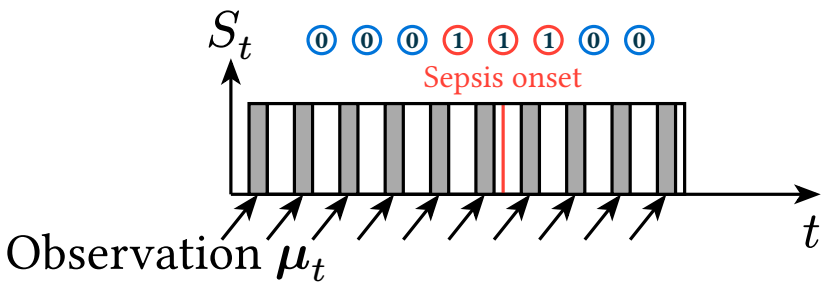

$\boldsymbol{\mu}_t$

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Sepsis Risk
$\tilde{S}_t = \tilde{A}_t \tilde{I}_t$

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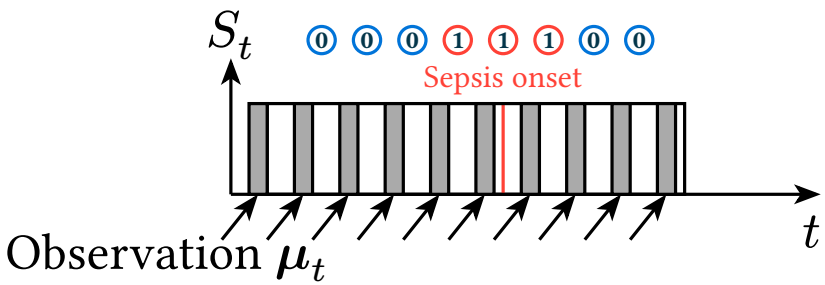

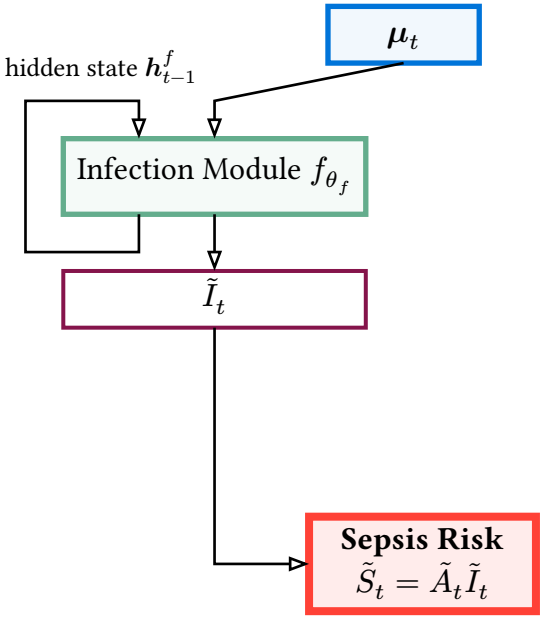

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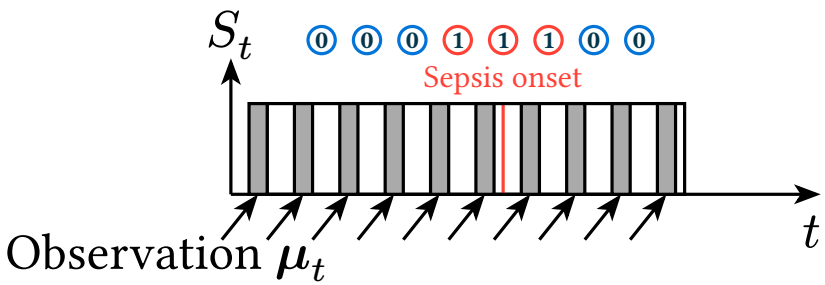

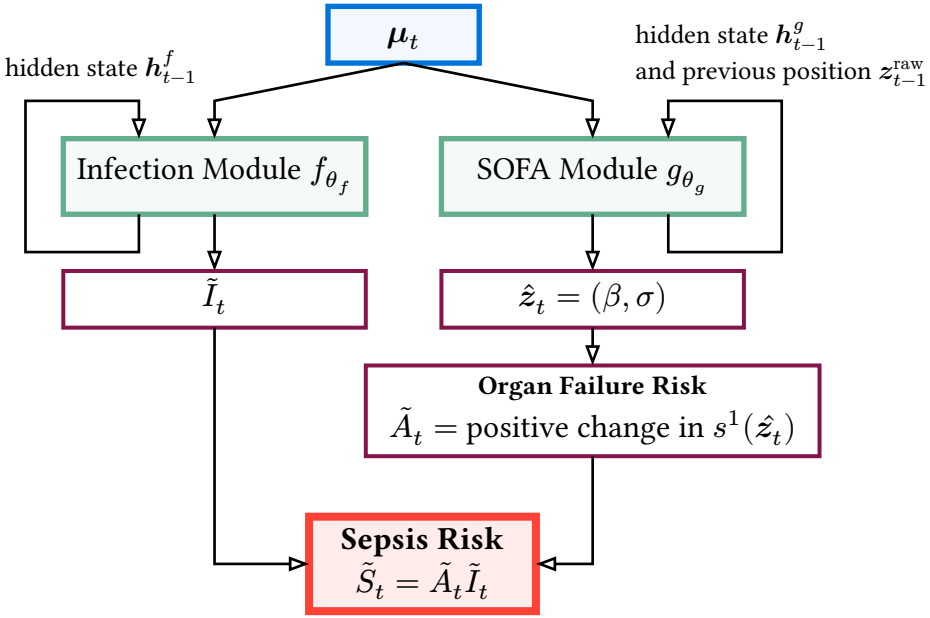

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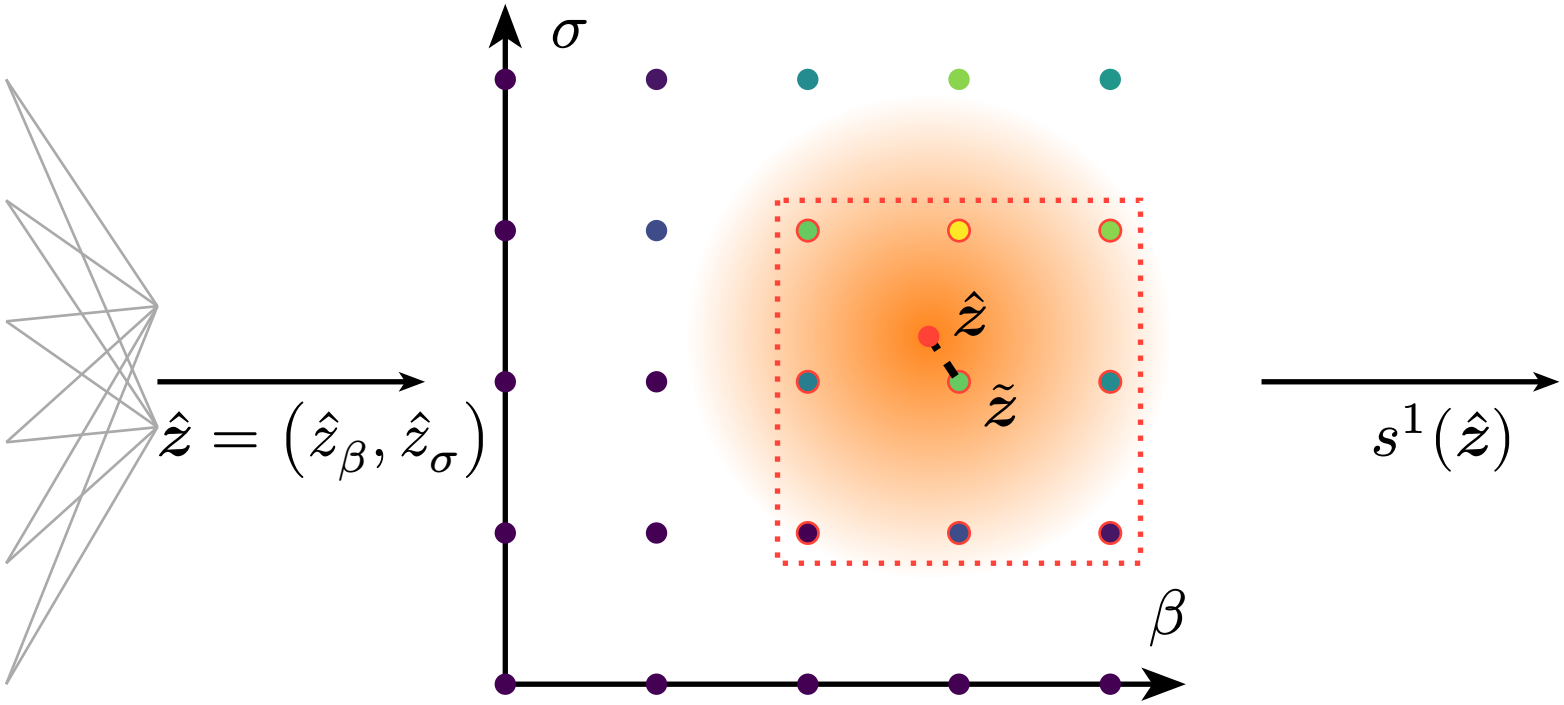

Experimental Results

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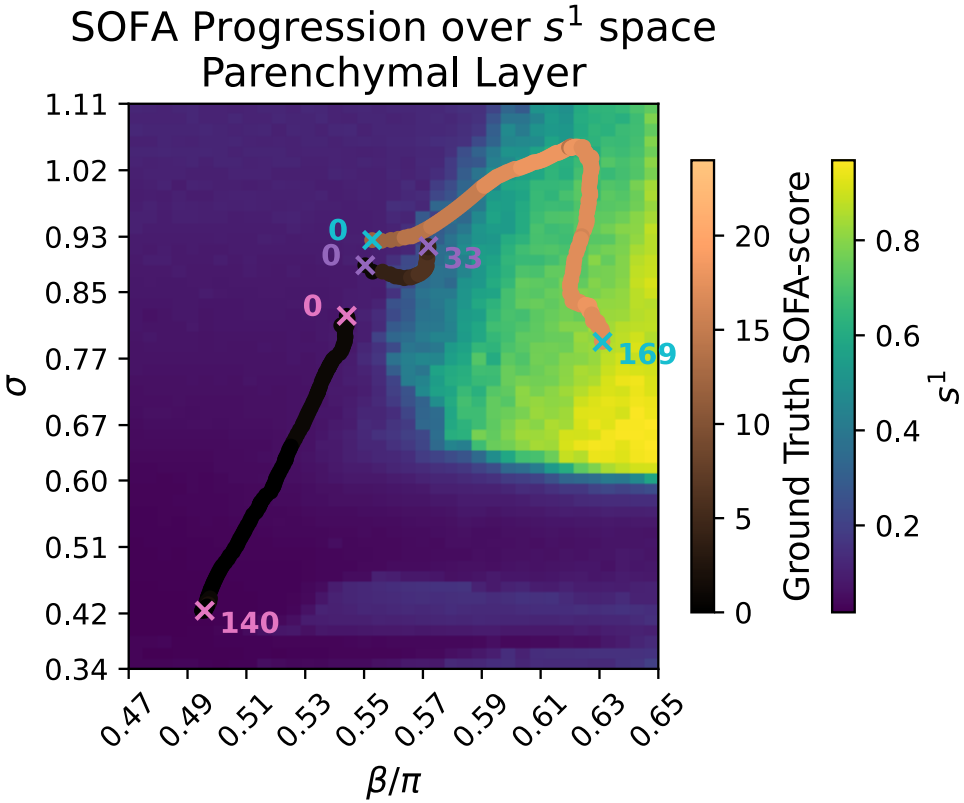

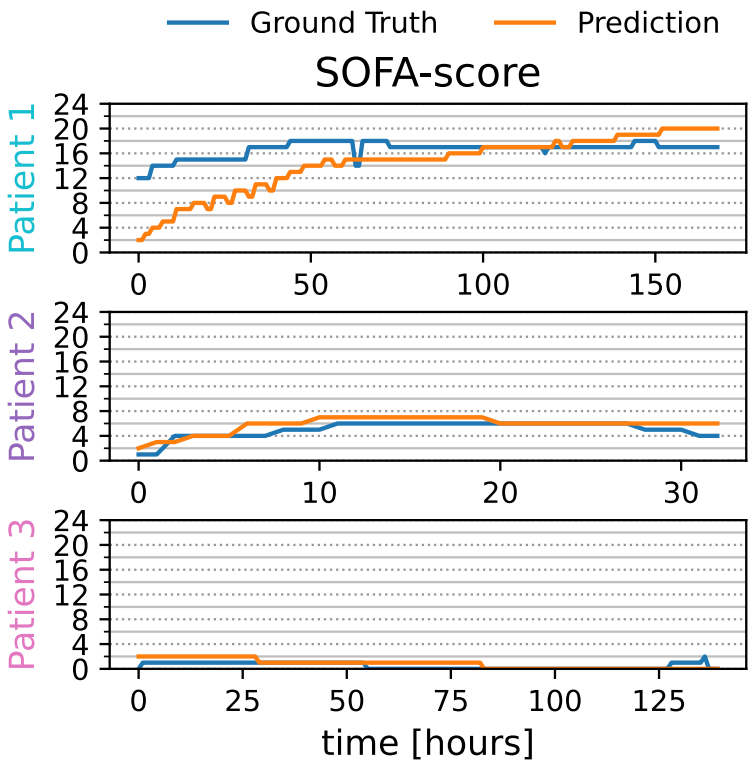

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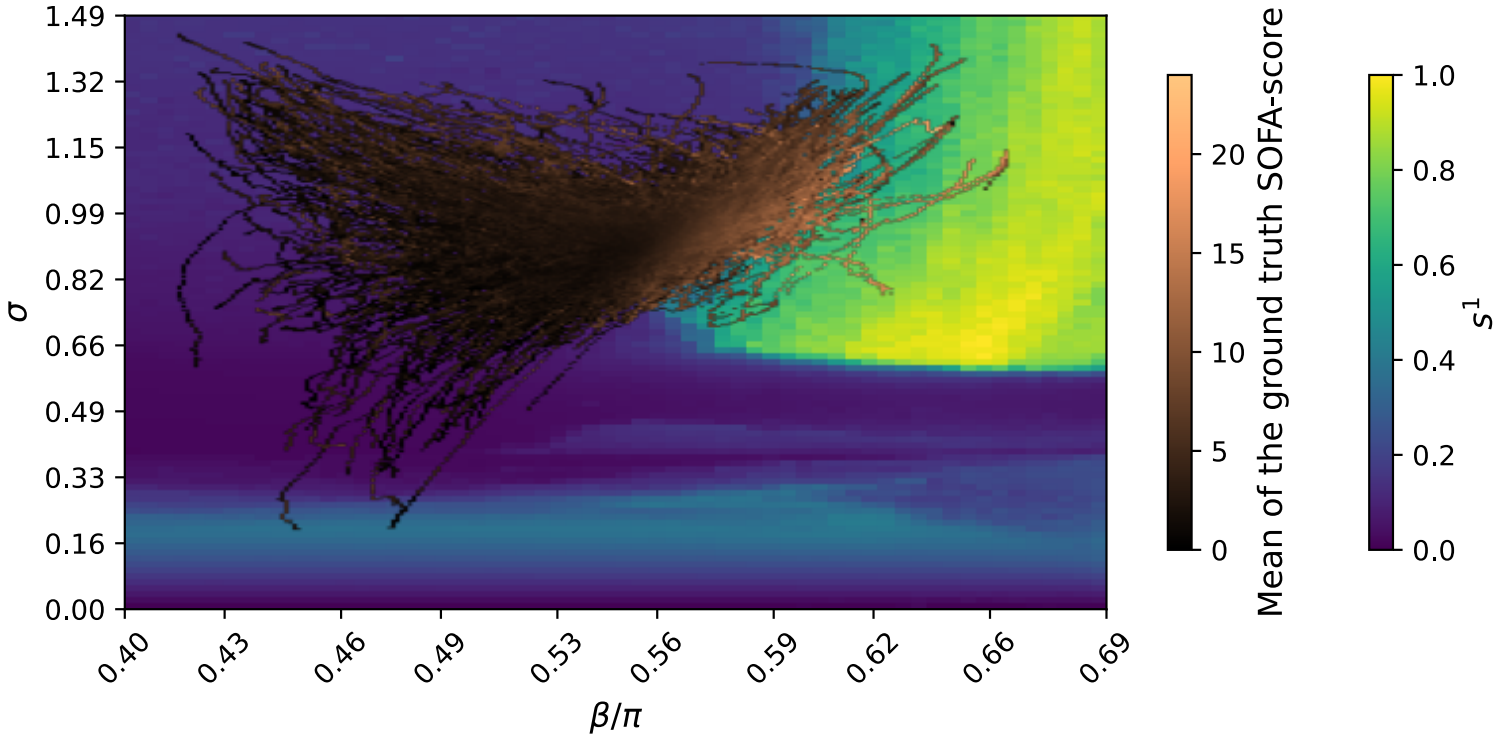

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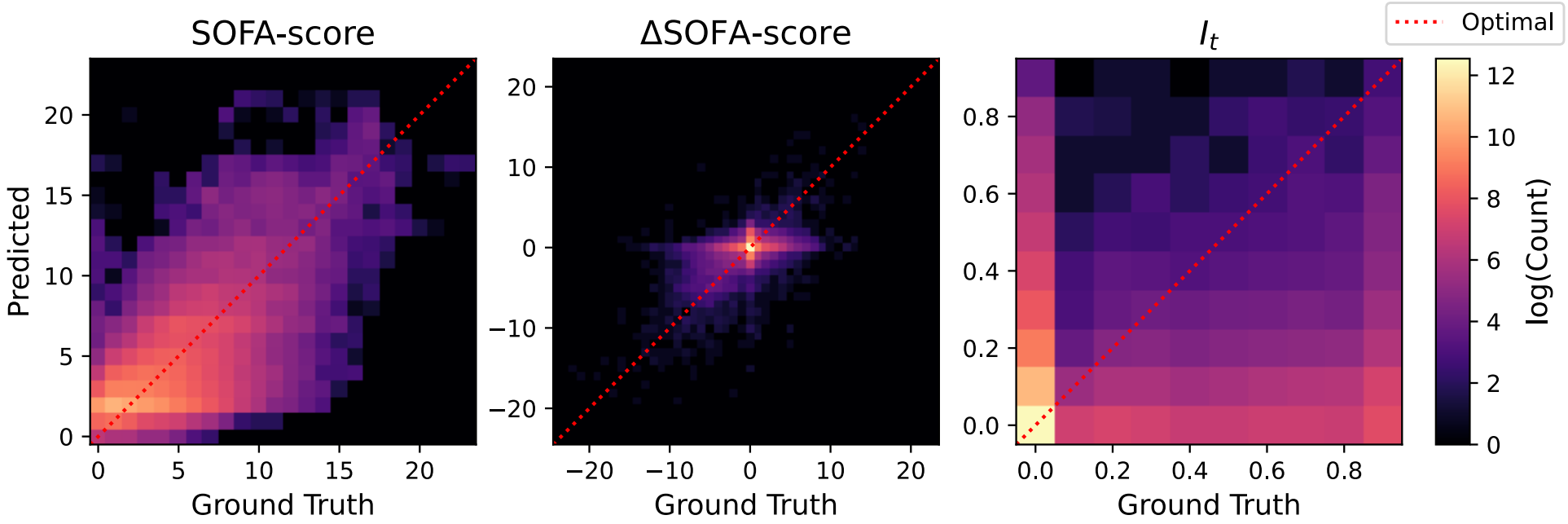

Experimental Results

Comparison against mean performance of YAIB [2], AUROC × 100 (↑, higher is better) and AUPRC × 100 (↑).

Model	**AUROC**	**AUPRC**
YAIB		
Regularized Logistic Regression	77.1	4.6
Light Gradient Boosting Machine	77.5	5.9
Transformer	80.0	6.6
LSTM	82.0	8.0
Temporal Convolutional Network	82.7	8.8
GRU	**83.6**	**9.1**
This work		
LDM	**84.69**	**9.87**

Conclusion

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Strengths

- Interpretable
- Decomposition Strategy
- Competitive Performance

Limitations and Outlook

- Missing external validation
- No ablation study for the LDM
  1. What parts are actually necessary?
  2. How much does the DNM contribute, do other sepsis models exist?
- Backpropagate through DNM?

Bibliography

[1] D. H. P. Directors, “Stroma and Parenchyma.” [Online]. Available: https://digitalhistology.org/organs-systems/general-concepts/stroma-and-parenchyma/stroma-and-parenchyma-1/

[2] R. van de Water, H. N. A. Schmidt, P. Elbers, P. Thoral, B. Arnrich, and P. Rockenschaub, “Yet Another ICU Benchmark: A Flexible Multi-Center Framework for Clinical ML,” in *The Twelfth International Conference on Learning Representations*, Oct. 2024.

[3] N. Bennett, D. Plečko, and I.-F. Ukor, “Sepsis 3 label — sep3.” [Online]. Available: https://eth-mds.github.io/ricu/reference/label_sep3.html

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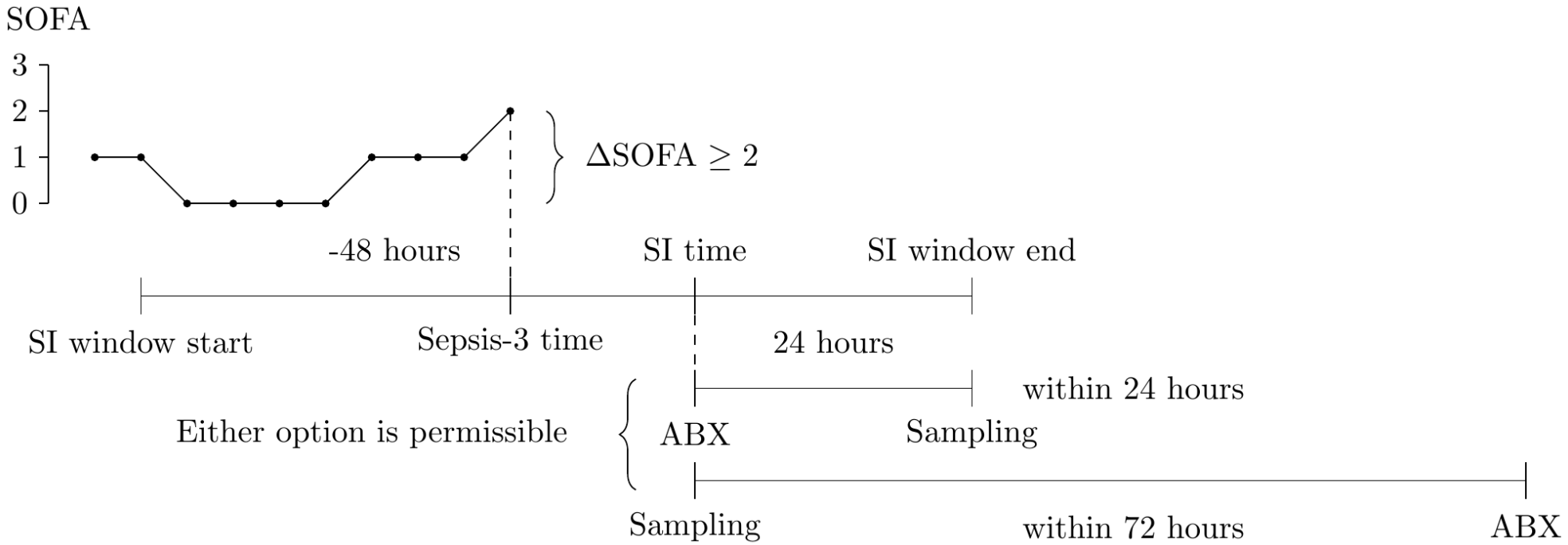
Graphical representation of the timings in the Sepsis-3 definition, taken from [3].



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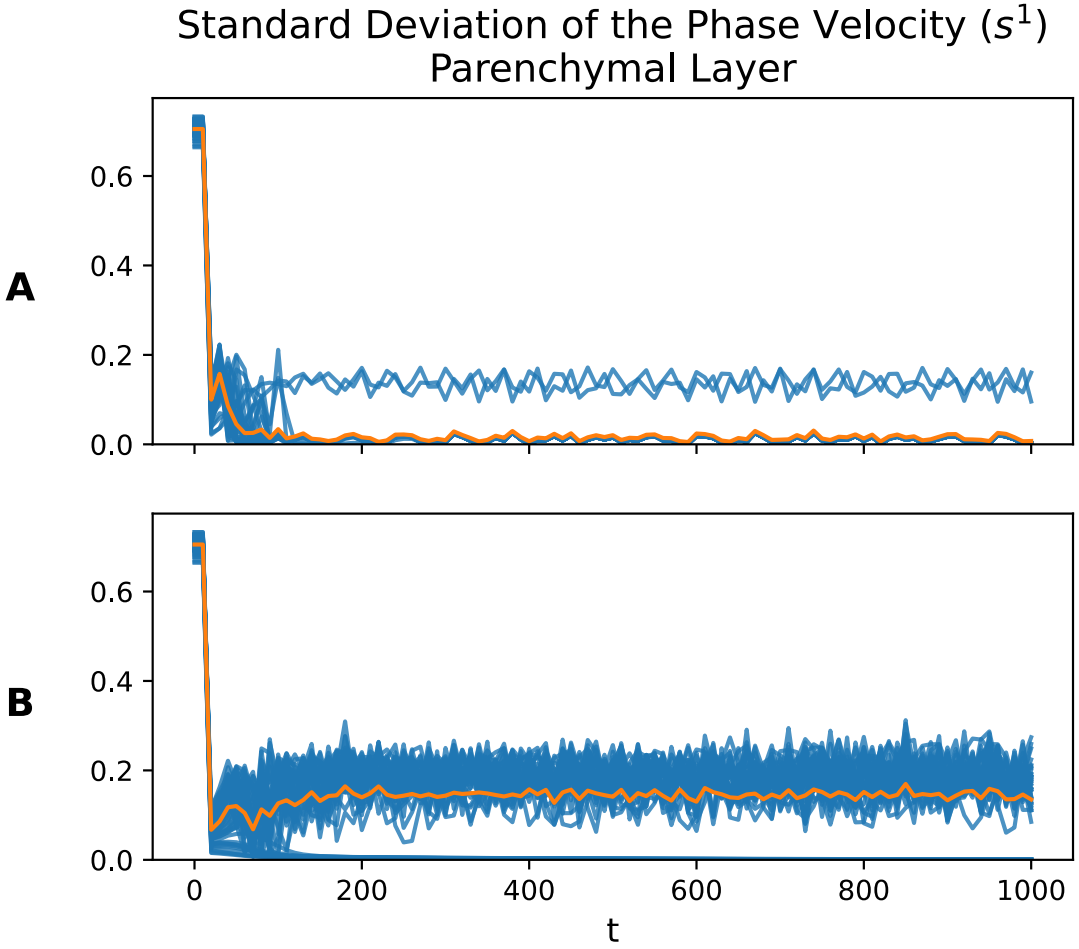

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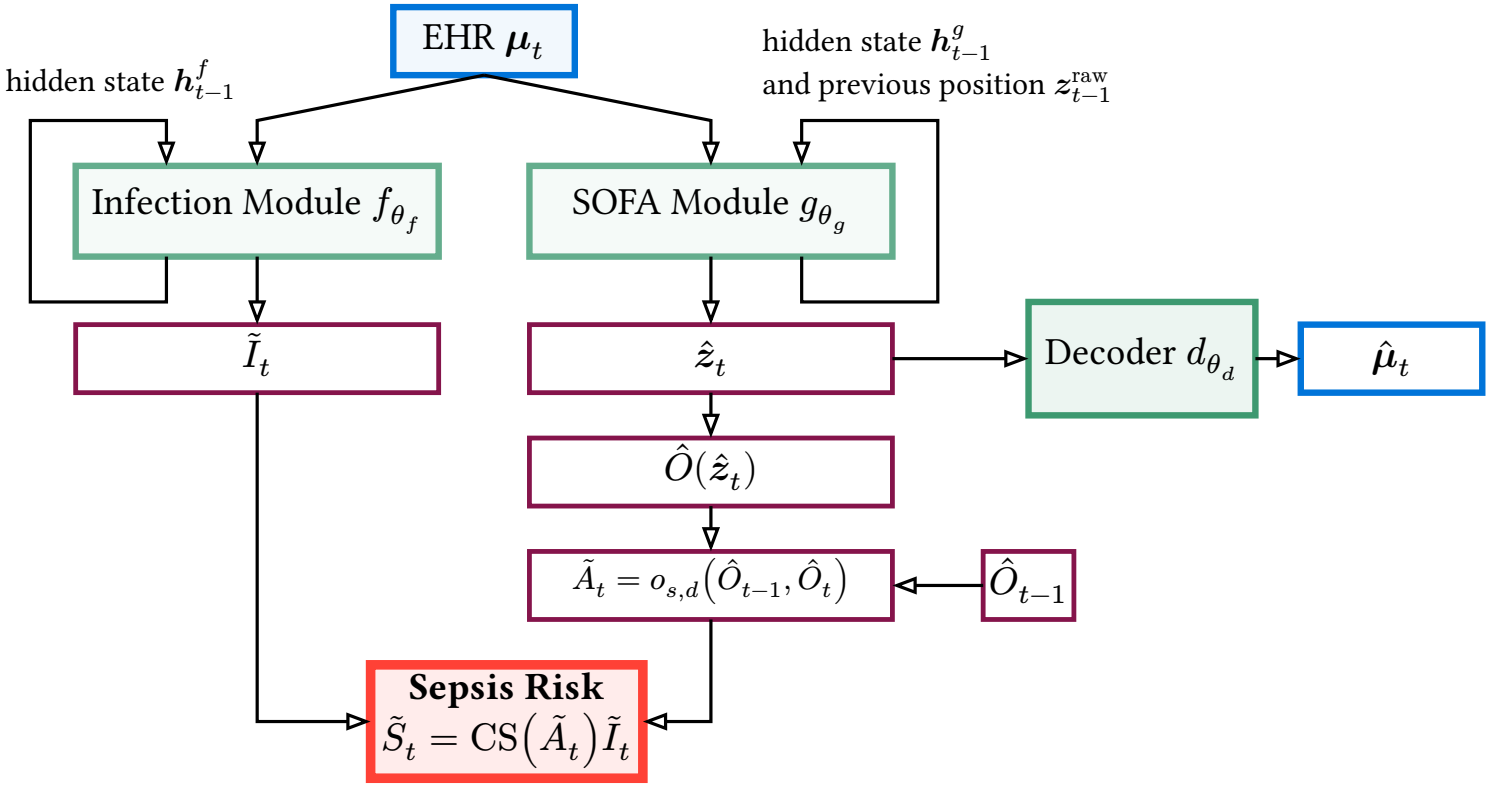

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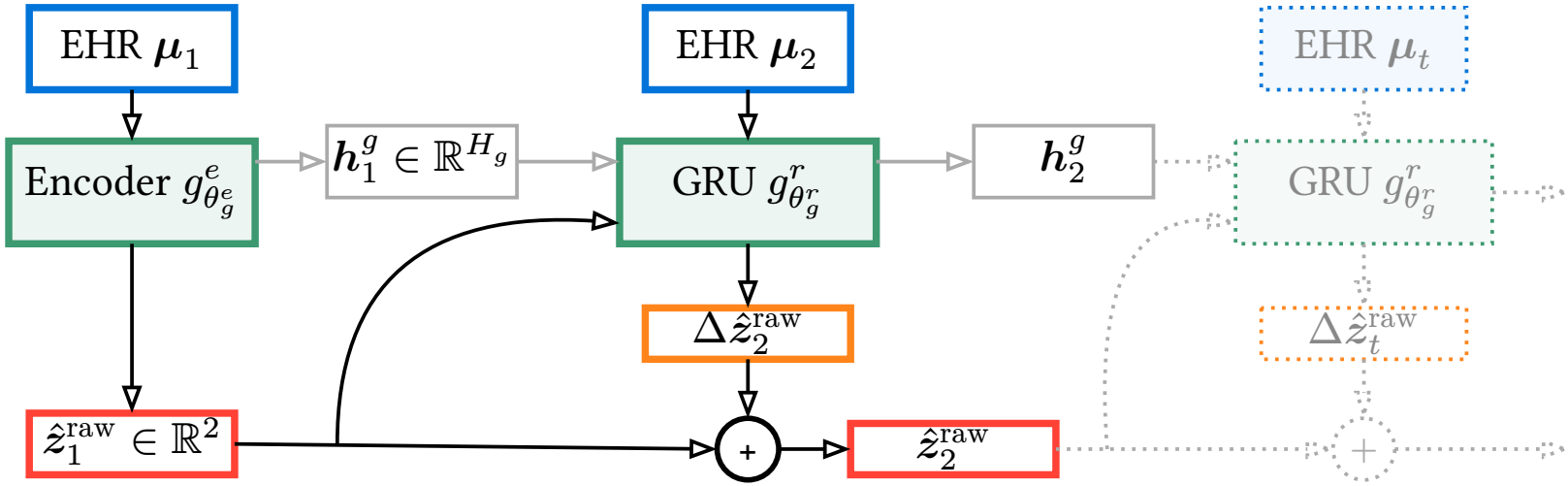

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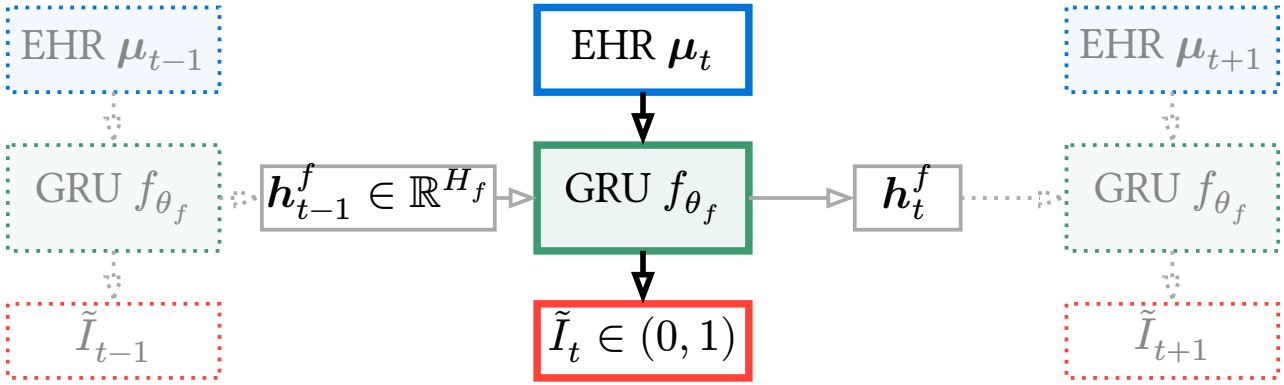

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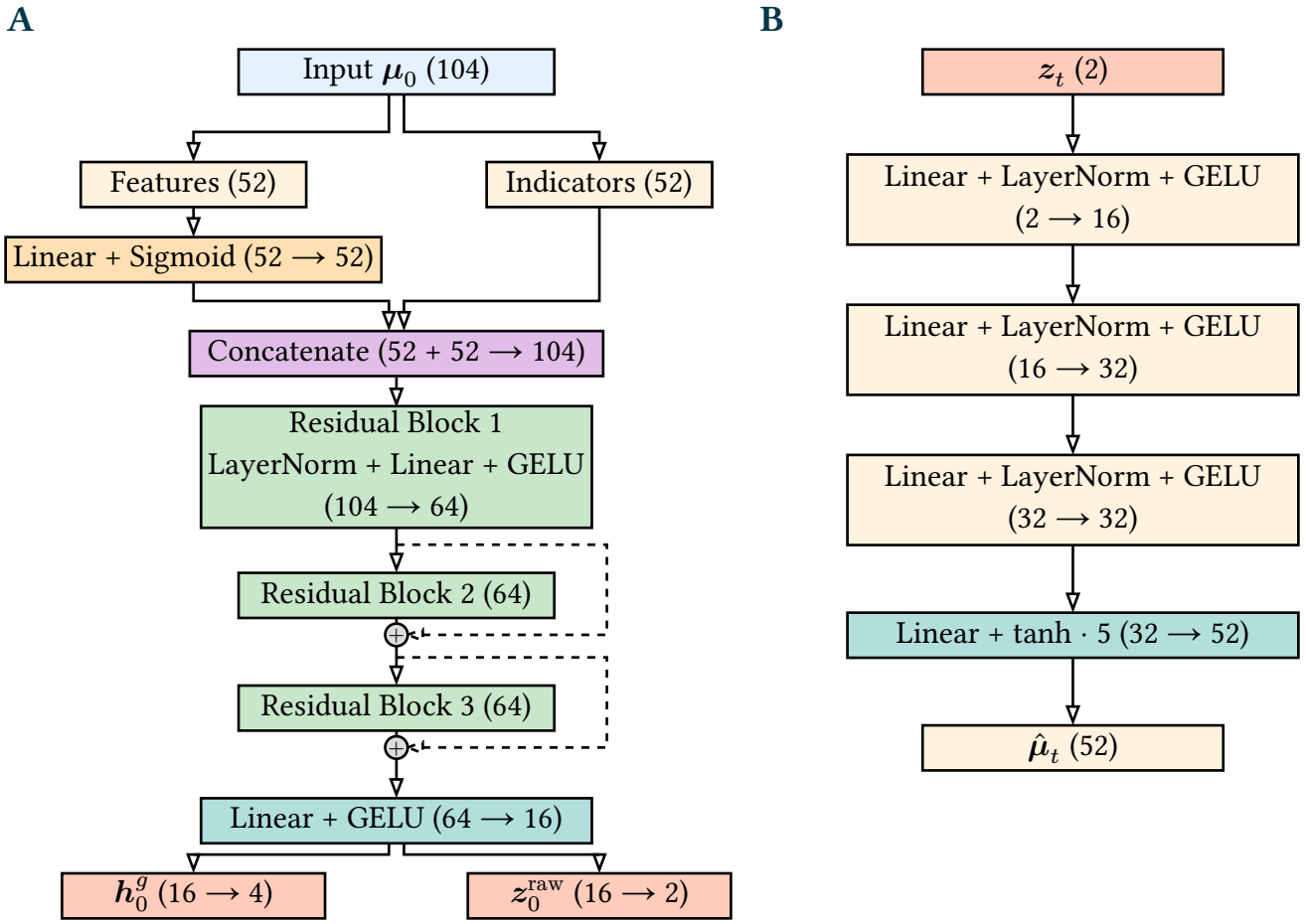

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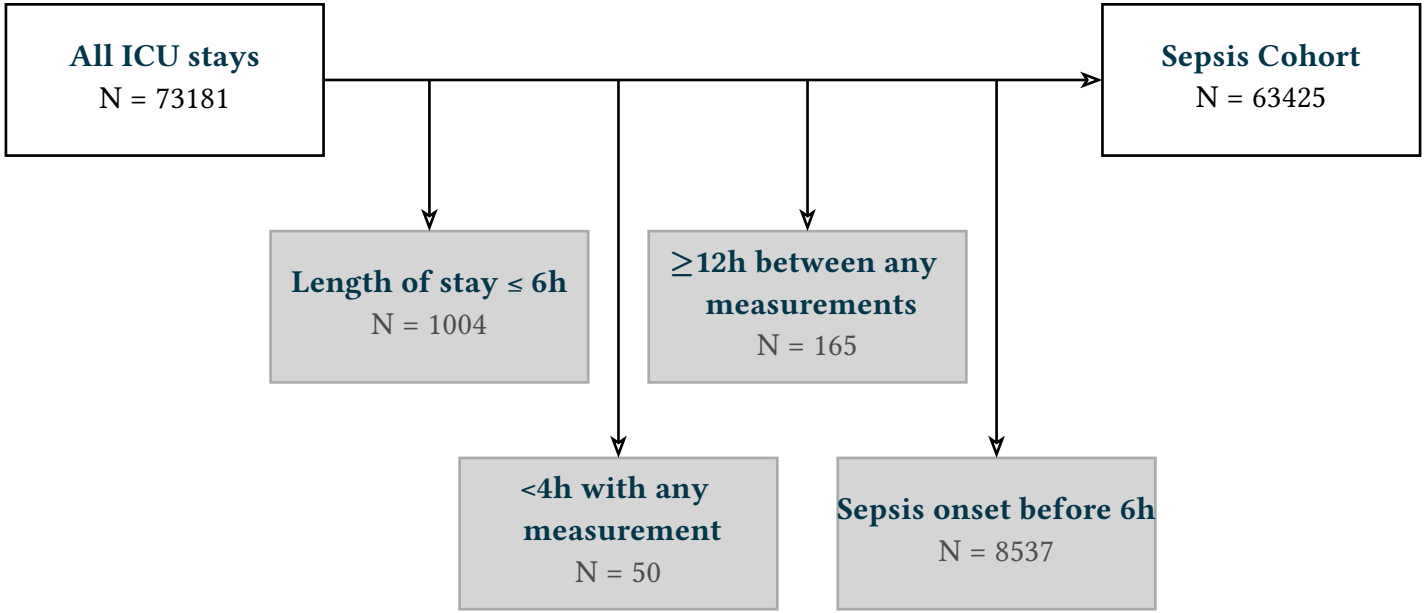

Bibliography

Characteristic	All patients	SEP-3 positive	SEP-3 negative
Demographics			
N	63425 (100.0)	3320 (5.2)	60105 (94.8)
Male	35170 (55.5)	1881 (56.7)	33289 (55.4)
Age at admission	65.0 (53.0–76.0)	65.0 (54.0–76.0)	65.0 (53.0–76.0)
Weight at admission	77.6 (65.1–92.3)	77.6 (65.6–94.0)	77.6 (65.0–92.2)

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Characteristic	All patients	SEP-3 positive	SEP-3 negative
Clinical Outcomes			
SOFA median	3.0 (1.0–5.0)	3.0 (1.0–5.0)	3.0 (1.0–5.0)
SOFA max	4.0 (2.0–6.0)	5.0 (4.0–8.0)	4.0 (2.0–6.0)
Hospital LOS hours	157.7 (92.8–268.9)	335.1 (194.2–548.6)	150.3 (90.9–256.0)
Hospital Mortality	4828 (7.6)	879 (26.5)	3949 (6.6)
Sepsis-3 onset time	-	13.0 (8.0–34.0)	-

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Characteristic	All patients	SEP-3 positive	SEP-3 negative
Ethnicity			
White	40364 (63.6)	2087 (62.9)	38277 (63.7)
Black	5809 (9.2)	262 (7.9)	5547 (9.2)
Asian	721 (1.1)	42 (1.3)	679 (1.1)
Hispanic	630 (1.0)	32 (1.0)	598 (1.0)
Other/Unknown	14924 (23.5)	897 (27.0)	14027 (23.3)

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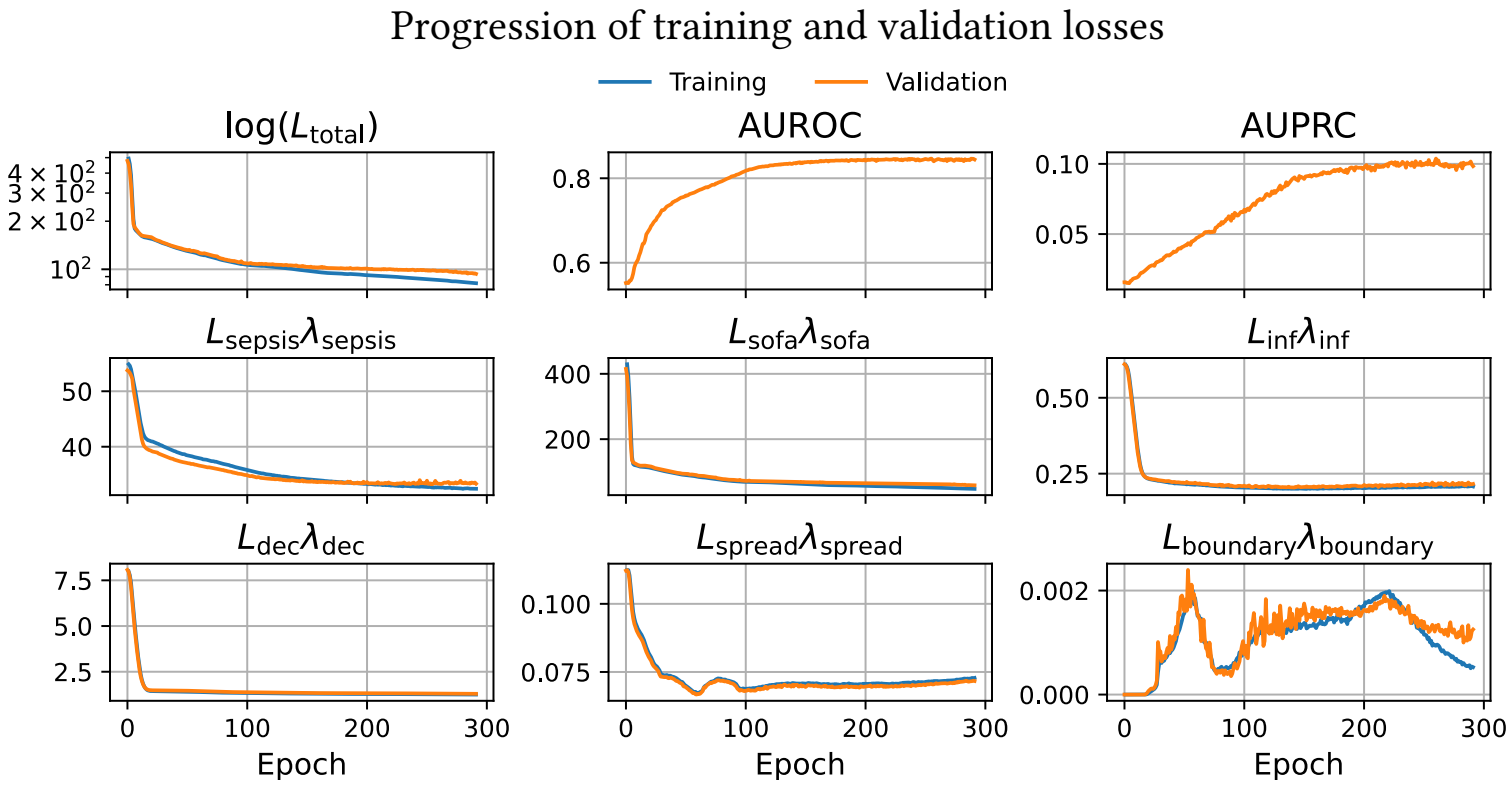
Characteristic	All patients	SEP-3 positive	SEP-3 negative
Admission Type			
Medical	45009 (71.0)	2817 (84.8)	42192 (70.2)
Surgical	2239 (3.5)	45 (1.4)	2194 (3.7)
Other/Unknown	15200 (24.0)	458 (13.8)	14742 (24.5)

Bibliography

Overview of loss components in the LDM training objective.

Loss	Type	Purpose	Supervises
$\mathcal{L}_{\text{sepsis}}$	BCE	Primary sepsis prediction	$f_{\theta_f}, g^e_{\theta^e_g}, g^r_{\theta^r_g}$
$\mathcal{L}_{\text{sofa}}$	Weighted MSE	SOFA estimation	$g^e_{\theta^e_g}, g^r_{\theta^r_g}$
$\mathcal{L}_{\text{inf}}$	BCE	Infection indicator	f_{θ_f}
$\mathcal{L}_{\text{spread}}$	Covariance	Latent diversity	$g^e_{\theta^e_g}, g^r_{\theta^r_g}$
$\mathcal{L}_{\text{boundary}}$	Positional	Latent Space	$g^e_{\theta^e_g}, g^r_{\theta^r_g}$
$\mathcal{L}_{\text{dec}}$	MSE	Latent semantics	$d_{\theta_d}, (g^e_{\theta^e_g}, g^r_{\theta^r_g})$

Bibliography



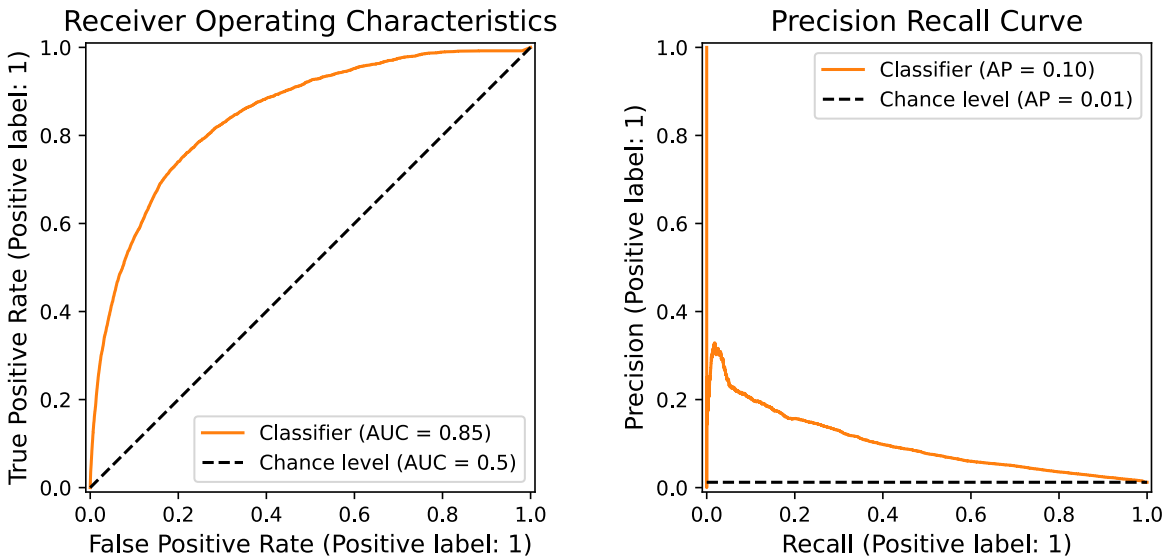
Bibliography

$$\delta\left(\tilde{S}_t\right) = \mathbb{I}\left(\tilde{S}_t > \tau\right)$$

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{TP} + \mathrm{FN}}$$

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

Receiver Operating Characteristic and Precision-Recall Curve.